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How is GenAI used in language classes:

Constructing a survey for language teachers

The TPACK Framework & Context

Motivation

- ✓ We need to **define** what **language teachers** need to **know** to **use AI** in their teaching
- ✓ We are **expanding** an existing framework, **TPACK**, to help **define** what **language teachers need to know** about **technology** to teach
- ✓ Our **goal** is to provide **guidance** on creating **professional development materials** for language teachers

Uncovering Teacher Knowledge

- ✱ The traditional **TPACK** (Technological Pedagogical Content Knowledge) model shows that **teaching requires** more than **knowing how to** operate a device.
- ✱ It involves **balancing** what is taught (**Content**), how it is taught (**Pedagogy**), and the tools used (**Technology**). In our case, the content is **language**, and the technology is **GenAI**.

Survey Objectives & Design Focus

Context of Participants

Language professionals (**teachers** and **teacher educators**) around the world, at all levels, who teach languages from K-12 through post-secondary education.

AI-TPACK Construct

AI Technical Skills (AI TK)

Evaluates the **technical competence** required to **access, prompt, and execute** activities.

AI Language Capabilities (AI TCK)

Measures a **teacher's knowledge** of what AI can actually do with **language**.

AI in Language Teaching (AI TPCK)

Assesses the **pedagogical knowledge** needed to combine **AI tools** with **teaching** methods effectively.

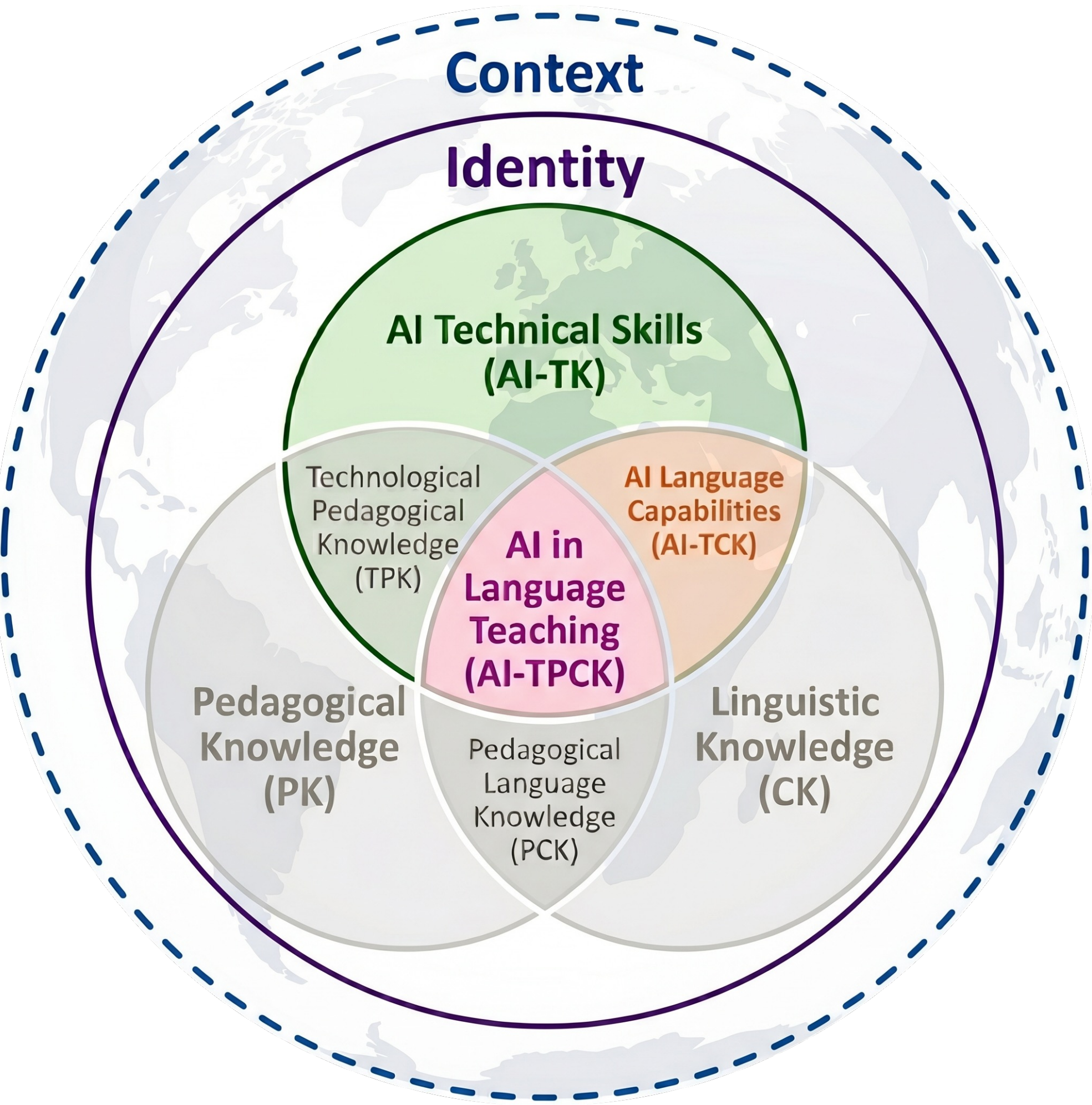
Professional Identity

Captures the educator's **internal shift**. It **measures** how **AI** **reshapes** a teacher's **professional self-image**

Local Context

Measures how educators navigate **technological constraints**, respect **local cultural norms**, and adapt to specific **institutional policies** regarding **AI use**.

RQ: How can we develop a valid survey instrument to assess language teachers' AI-TPACK?



Adapted from Mishra (2019) and Visoná et al. (2025)

Step 1: Domain Scoping & TLU Analysis

3 expert linguists reviewed **42 articles** about TPACK and mapped Target Language Use (TLU) from **33 empirical studies** to **identify** real-world **tasks** used with **GenAI** by **language teachers**.

Step 2: Authentic Items

Interviews with **15 ESL/EFL teachers** captured **real-world AI challenges** and **practices**.

Step 3: Categorizing 5 Constructs

Based on the literature, TLU mapping, and interview feedback, the 3 experts drafted an initial pool of **86 survey items**. These were categorized into **5 constructs**: **AI-TK**, **AI-TCK**, **AI-TPCK**, **Identity**, and **Context**.

Step 4: Expert Content Validation

15 assessment and teaching experts provided **88 comments** to remove jargon, item nesting, and ensure accessibility for **language educators worldwide**.

Step 5: Refinement & Implementation

3 expert linguists revised down the survey to **74 items** on a **1–5 Likert scale** all reviewed by the initial **15 teacher** interviewed.

Survey Items



Take the survey

12:29

Q1. I understand the pedagogical contribution of AI tools to my language teaching.

Strongly disagree

Disagree

Undecided

Agree

Strongly agree

12:29

Q43. I can record my video or speech, upload it to an AI-pronunciation service, and download the diagnostic feedback report.

Strongly disagree

Disagree

Undecided

Agree

Strongly agree

Next Steps

Global Distribution

Distributing the 74-item **survey** to **over 1,750 participants worldwide** to include a wide variety of teachers and teaching contexts.

Statistical Validation

Data will undergo **item analysis** and **Confirmatory Factor Analysis** to assess the **quality of measurement** and make necessary revisions

Results

The goal is to translate our findings into **context-sensitive strategies for teachers**, provide the **AI-TPACK survey** instrument to **language professionals**, and create **professional development material** for the field.

References

Mishra, P. (2019). Considering contextual knowledge: The TPACK diagram gets an upgrade. Journal of digital learning in teacher education, 35(2), 76-78.

Visoná, M. W., Chapelle, C. A., Kurt, S., Yang, J., Guskarska, A., & Bordbarjavid, F. (2025). Gathering Evidence of Teachers' TPACK through Discourse Analysis of Their Coursework in an Online Professional Development Course: A Case Study. In Handbook of Technological Pedagogical Content Knowledge (TPACK) for Educators (pp. 373-400). Routledge.